

Project Report: Voice-Based Cognitive Decline Pattern Detection

1. Introduction

This project builds a **proof-of-concept pipeline** for detecting **cognitive decline patterns** based on **voice recordings**. It leverages **audio features** and **linguistic content** to detect speech-related patterns that could hint at early signs of cognitive impairment.

2. Methodology and Code Explanation

a) Preprocessing and Transcription

- The uploaded audio files are transcribed into text using **Google's Speech Recognition API**.
- Audio files are processed using librosa, a popular audio analysis library.

Code Snippet:

```
recognizer = sr.Recognizer()
with sr.AudioFile(audio_path) as source:
    audio = recognizer.record(source)
transcript = recognizer.recognize_google(audio)
```

b) Feature Extraction

From Audio:

- **Duration:** Total length of the audio.
- **Tempo:** Speed of speech (beats per minute estimated).
- **Pitch Features:**
 - **Mean Pitch:** Average frequency, indicating tone variation.
 - **Pitch Variability:** Standard deviation of pitch, indicating monotony or expressiveness.
- **Pauses:** Silent segments are extracted to find gaps in speech (proxy for hesitations).

From Text (Transcript):

- **Word Count:** Number of spoken words.
- **Lexical Diversity:** Vocabulary richness = (Unique words) / (Total words).
- **Sentiment Score:** Emotional tone using a **Huggingface transformer model**.

Code Example for Audio Feature Extraction:

```
duration = librosa.get_duration(y=y, sr=sr)
tempo, _ = librosa.beat.beat_track(y=y, sr=sr)
pitches, magnitudes = librosa.piptrack(y=y, sr=sr)
mean_pitch = np.mean(pitches[pitches > 0])
pitch_variability = np.std(pitches[pitches > 0])
```

c) Clustering (Unsupervised Learning)

- **Algorithm Used:** KMeans
- **Why KMeans?**
 - Since labels (cognitive impairment vs. normal) were **not pre-provided**, unsupervised clustering is ideal.
 - KMeans helps find **natural groupings** based on the extracted features.
- **Number of Clusters:** Typically set manually (e.g., n_clusters=2 or n_clusters=3) to observe grouping.

Code Example:

```
kmeans = KMeans(n_clusters=2, random_state=0)
kmeans.fit(feature_matrix)
```

Visualization:

- Feature clusters are plotted using PCA-reduced 2D space for interpretation.
-

3. Results and Interpretation

- **Clusters formed** based on speech fluency, tempo, pitch variability, and sentiment scores.
 - Potential observations:
 - **Low lexical diversity, more pauses, lower tempo** → Might cluster into **high cognitive decline risk** group.
 - **Higher lexical richness, stable pitch, fast tempo** → Might form a **healthy cognition** group.
 - **Cluster centroids** showed distinct feature patterns, indicating good separation even without labels.
-

4. Insights: Which Features Were Most Insightful?

The **most impactful features** for clustering were:

Feature	Why Important for Cognitive Detection
Lexical Diversity	Lower diversity indicates vocabulary loss.
Pitch Variability	Monotone speech often linked with neurological disorders.
Pauses (Speech Gaps)	Hesitations and long pauses suggest cognitive effort.
Tempo	Slowed speaking pace correlates with cognitive decline.
Sentiment Polarity	Emotional flatness (low sentiment variance) can indicate mental decline.

5. ML Methods Used and Why

Method	Reason
KMeans Clustering	No labels were available, so unsupervised discovery of patterns was ideal.
Speech Recognition (Google API)	Quick and robust way to extract text from voice without training a model.

Method	Reason
Sentiment Analysis (Transformers)	Provides an emotional dimension to complement cognitive features.

6. Potential Next Steps to Make it Clinically Robust

Next Step	Description
Use a larger and diverse dataset	Clinical studies need statistically significant data from varied demographics.
Supervised Learning Models	Once labeled data (diagnoses) are available, models like SVM, Random Forests, or Deep Learning (e.g., LSTM) can be trained.
Longitudinal Analysis	Study changes over time rather than just one-off recordings.
Incorporate Medical Annotations	Clinician annotations on speech errors, coherence loss, and emotional flattening can enhance model features.
Robust Noise Handling	Real-world voice samples often have noise; improve preprocessing robustness.
Feature Engineering on Prosody	Include features like speaking rate, articulation rate, jitter, shimmer, etc.
Explainable AI (XAI)	Build interpretable models that clinicians can trust.

7. Conclusion

This project successfully demonstrates that **audio and text features** can provide meaningful signals for detecting **cognitive decline**.

While preliminary, the **MemoTag** pipeline lays a **strong foundation** for future clinical-grade, automated cognitive assessment tools.